Light

Memory & Storage

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Assessment Hub | Notes / Textbook | Long-Answer Practice

Assessment Hub
Long-Answer
Notes Index
What you must know
Toolkit
1. Primary memory
2. Secondary storage
3. Choosing storage
Worked examples
Quick reference
Typical pitfalls

Edexcel GCSE Computer Science — **Memory and Storage (1CP2/01 Topic 3: Computers)**. This topic covers the differences between primary memory (RAM and ROM) and secondary storage, the characteristics of magnetic, optical and solid-state storage technologies, and how to select appropriate storage for different purposes. Exam questions typically ask you to **state** the purpose of RAM and ROM, **describe** characteristics of storage types, **explain** why one storage type is more suitable than another, and **calculate** storage requirements. The 2024 Paper 1 included questions on storage characteristics and file size calculations. Expect questions worth 2—6 marks that require you to compare storage technologies and justify choices.

Why memory and storage matters

Every computer system needs both memory (for immediate processing) and storage (for keeping data permanently). Understanding the difference helps you:

- **Understand system performance:** RAM affects how many programs can run simultaneously and how responsive the system feels
- **Make informed hardware choices:** Knowing SSD vs HDD trade-offs helps when selecting or upgrading computers

- **Understand data persistence:** Why unsaved work is lost when power fails, but saved files remain
- **Calculate storage needs:** Working out how much space files require and choosing appropriate media
- **Succeed in the exam:** Memory and storage appears in Paper 1 regularly, typically worth 4—10 marks across multiple questions
  In the exam, you will be asked to compare storage types, explain why certain characteristics matter for specific uses, and perform calculations involving file sizes and storage capacity.

What you must know

This checklist covers the key exam-visible concepts from the Edexcel specification (3.1.2). You should be able to explain each point and apply it to exam scenarios.

- **Define** what is meant by primary memory and explain the difference between RAM and ROM
- **Describe** the purpose of RAM and ROM in a computer system and state which is volatile
- **Explain** why secondary storage is needed in addition to primary memory
- **Describe** the characteristics of magnetic, optical and solid-state storage technologies
- **Compare** storage types using characteristics: capacity, speed, portability, durability, reliability and cost
- **Calculate** file sizes and storage requirements using appropriate units (bytes, KiB, MiB, GiB, TiB)

> **Specification link:** This topic covers Edexcel 1CP2 specification section 3.1.2 — "understand the need for primary storage (main memory) and secondary storage, explaining the differences between RAM and ROM" and 3.1.3 — "understand the different types of secondary storage (solid-state, optical and magnetic), explaining the operation of solid-state storage and giving examples of where each type of storage might be used."

Toolkit

These are the key patterns, definitions and relationships you need to apply in exam questions.

Memory hierarchy

```
┌──────────────────────────────────────────────────────────────────────────────┐ │ MEMORY
HIERARCHY │ │ │ │ FASTEST ──────────────────────────────────────────────────────────► SLOWEST │
│ SMALLEST ──────────────────────────────────────────────► LARGEST │ │ MOST
EXPENSIVE ──────────────────────────────────────────► CHEAPEST │ │ │ │ ┌──────────┐
┌──────────┐ ┌──────────────────────────┐ │ │ │ Registers│ → │ Cache │ → │
RAM │ │ │ │ (CPU) │ │ (CPU) │ │ (Primary memory) │ │ │ └──────────┘
└──────────┘ └──────────────────────────┘ │ │ │ │ │ │
└──────────┴──────────────────────┘ │ │ │ │ ▼ │ │
┌──────────────────────────────────────────────────────────────────────────┐ │ │ │ SECONDARY
STORAGE │ │ │ │ SSD (solid-state) │ HDD (magnetic) │ Optical │ │ │ │ Flash
drives │ Tape │ CD/DVD/Blu-ray│ │ │
└──────────────────────────────────────────────────────────────────────────┘ │ │ │ │ PRIMARY
= Volatile (loses data without power) + Fast │ │ SECONDARY = Non-volatile
```

```
(keeps data without power) + Slower │
└──────────────────────────────────────────────────────────────────────┘
```

Storage unit conversions

Unit	Symbol	Size in bytes	Power of 2	Conversion
Byte	B	1	2^0	8 bits = 1 byte
Kibibyte	KiB	1,024	2^{10}	1,024 bytes = 1 KiB
Mebibyte	MiB	1,048,576	2^{20}	1,024 KiB = 1 MiB
Gibibyte	GiB	1,073,741,824	2^{30}	1,024 MiB = 1 GiB
Tebibyte	TiB	1,099,511,627,776	2^{40}	1,024 GiB = 1 TiB

Exam tip: Edexcel uses binary prefixes (KiB, MiB, GiB) based on powers of 1024, not decimal prefixes (KB, MB, GB) based on powers of 1000. Always multiply or divide by 1024 when converting between units.

Storage characteristics comparison

Characteristic	SSD (solid-state)	HDD (magnetic)	Optical (CD/DVD/Blu-ray)
Capacity	Up to several TiB	Up to 20+ TiB	700 MiB — 128 GiB
Speed	Very fast (no moving parts)	Slower (mechanical read/write head)	Slowest (spinning disc + laser)
Durability	High (no moving parts, shock resistant)	Low (fragile, sensitive to movement)	Medium (can be scratched)
Portability	High (small, lightweight)	Medium (heavier, fragile)	High (thin, lightweight discs)
Cost per GiB	Higher	Lower	Very low (for read-only media)
Reliability	High (limited write cycles)	Medium (mechanical failure risk)	Medium (degradation over time)

Key formulas

File size calculation:

File size (bytes) = number of characters × bytes per character (for text files)
File size (bytes) = resolution × colour depth ÷ 8 (for images)
File size (bytes) = sample rate × duration × bit depth ÷ 8 (for audio)
Unit conversion:
To convert bytes → KiB: divide by 1024
To convert KiB → MiB: divide by 1024
To convert MiB → GiB: divide by 1024

1. Primary memory

Primary memory (also called main memory) is the memory that the CPU can access directly. It holds the data and instructions that are currently being used. Primary memory is essential for a computer to function — without it, the CPU would have nowhere to store the programs it is running.

1.1 RAM (Random Access Memory)

RAM is the main working memory of a computer. When you open a program or file, it is loaded from secondary storage into RAM so the CPU can access it quickly.

Key characteristics of RAM:

- **Volatile:** RAM loses all its contents when the power is turned off — this is why unsaved work is lost if your computer crashes
- **Read and write:** Data can be read from RAM and written to RAM
- **Fast access:** Much faster than secondary storage, allowing the CPU to fetch data quickly
- **Temporary:** Holds data and instructions only while they are being used

What RAM stores:

- The operating system (while the computer is running)
- Currently running applications
- Data being processed (documents, images, etc.)

> **Exam tip:** The key word for RAM is **volatile**. If a question asks "What happens to data in RAM when the power is turned off?", the answer is: it is lost/erased/deleted.

1.2 ROM (Read-Only Memory)

ROM is memory that contains permanent instructions that do not change. The most important use of ROM is to store the **BIOS/UEFI** (Basic Input/Output System) — the firmware that starts up the computer.

Key characteristics of ROM:

- **Non-volatile:** ROM retains its contents even when the power is turned off
- **Read-only:** Data is written once (during manufacture) and cannot normally be changed by the user
- **Permanent:** Contents remain unchanged throughout the life of the computer
- **Small capacity:** Typically much smaller than RAM

What ROM stores:

- BIOS/UEFI — the boot-up instructions
- POST (Power-On Self-Test) routines
- Basic hardware configuration data

Boot process: When you turn on a computer, the CPU first runs the instructions stored in ROM. These instructions check the hardware (POST), then load the operating system from secondary storage into RAM.

1.3 Why do we need both RAM and ROM?

RAM and ROM serve different but complementary purposes. ROM provides the permanent instructions needed to start the computer, while RAM provides the temporary workspace for running programs.

Feature	RAM	ROM
Volatility	Volatile — loses data when power off	Non-volatile — keeps data when power off
Read/Write	Can read and write	Read-only (written during manufacture)
Purpose	Temporary storage for running programs	Permanent storage for boot instructions
Contents	OS, applications, user data (while in use)	BIOS/UEFI, POST routines
Typical size	4 GiB — 64 GiB (or more)	A few MiB

Worked Example 1 — Defining RAM and ROM

Question: State one difference between RAM and ROM. [2 marks]

Model answer:
RAM is volatile, meaning it loses its contents when the power is turned off **(1)**.
ROM is non-volatile, meaning it retains its contents when the power is turned off **(1)**.
Also acceptable: RAM can be written to, ROM is read-only / RAM stores running programs, ROM stores boot instructions

Exam tip: For "state one difference" questions, give a paired comparison — say what RAM does AND what ROM does. This ensures you get both marks.

Worked Example 2 — Calculating RAM requirements

Question: A computer runs an operating system that requires 2 GiB of RAM. The user wants to run three applications that each require 512 MiB of RAM. Calculate the minimum amount of RAM needed in GiB. [2 marks]

Step 1: Calculate total application RAM: 3 × 512 = 1536 MiB **(1)**
Step 2: Convert to GiB and add OS: 1536 ÷ 1024 = 1.5 GiB, so total = 2 + 1.5 = 3.5 GiB **(1)**
Also acceptable: 2048 + 1536 = 3584 MiB = 3.5 GiB

Exam tip: RAM questions often combine multiple requirements. Convert to the same unit first, then add. Remember: 1 GiB = 1024 MiB.

Worked Example 3 — Explaining volatility

Question: A user is typing a document but has not saved it. Explain why the document will be lost if the power fails. [2 marks]

Model answer:
The document is stored in RAM while being worked on **(1)**.
RAM is volatile, so when the power fails, all data in RAM is lost **(1)**.

Exam tip: This is a classic exam scenario. Link the situation (unsaved work) to the concept (volatility of RAM) to the consequence (data lost).

Worked Example 4 — Role of ROM in boot process

Question: Explain the role of ROM when a computer is first switched on. [2 marks]

Model answer:
ROM contains the BIOS/boot instructions that run when the computer is first switched on **(1)**.
These instructions perform hardware checks (POST) and then load the operating system from secondary storage into RAM **(1)**.

Exam tip: ROM's role is specifically about **starting up** the computer. Mention BIOS, POST, or loading the OS to show you understand the boot process.

2. Secondary storage

Secondary storage is non-volatile storage that keeps data permanently, even when the power is turned off. Unlike RAM, secondary storage is used for long-term data storage. There are three main types: magnetic, optical, and solid-state.

2.1 Why do we need secondary storage?

Primary memory (RAM) is volatile and has limited capacity. Secondary storage is essential because:

- **Non-volatile:** Data is retained when the computer is turned off
- **Large capacity:** Can store much more data than RAM (terabytes vs gigabytes)
- **Long-term storage:** Files, applications, and the operating system need permanent storage
- **Lower cost per byte:** Much cheaper than RAM for the same capacity

2.2 Magnetic storage

Magnetic storage uses magnetised patterns on a spinning disc or tape to store data. The most common example is the **Hard Disk Drive (HDD)**.

How HDDs work:

- Data is stored on spinning magnetic platters
- A read/write head moves across the platters to access data
- Data is stored as magnetic patterns representing 0s and 1s
- The platters spin at high speed (5400—7200 RPM typically)

Characteristics:

- **High capacity:** Up to 20+ TiB
- **Low cost per GiB:** Cheapest storage per byte
- **Slower access:** Mechanical parts cause delay (seek time)
- **Fragile:** Moving parts are sensitive to shocks and vibration
- **Noise:** Spinning platters and moving heads create noise

Typical uses: Desktop computers, servers, backup storage, data centres

> **Exam tip:** HDDs are best for **large capacity at low cost** where speed is less critical — like backup servers or storing large media libraries.

2.3 Solid-state storage

Solid-state storage uses flash memory with no moving parts. Examples include **SSDs** (Solid-State Drives), USB flash drives, and memory cards.

How SSDs work:

- Data is stored in flash memory chips (NAND cells)
- Electrical charges are trapped in cells to represent 0s and 1s
- No moving parts — data is accessed electronically
- Uses a controller chip to manage read/write operations

Characteristics:

- **Very fast:** No seek time, instant access
- **Durable:** No moving parts, shock resistant
- **Silent:** No noise from mechanical components
- **Low power:** Uses less electricity than HDDs
- **Higher cost per GiB:** More expensive than HDDs
- **Limited write cycles:** Flash memory degrades over many writes

Typical uses: Laptops, tablets, smartphones, gaming consoles, portable drives

> **Exam tip:** SSDs are best for **speed and durability** — ideal for laptops (portable, shock resistant) and systems where fast boot times matter.

2.4 Optical storage

Optical storage uses a laser to read and write data on a spinning disc. Examples include CDs, DVDs, and Blu-ray discs.

CD (Compact Disc)

Capacity: ~700 MiB
Uses: Music distribution, small software, archives
CD-R (write once), CD-RW (rewritable)

DVD (Digital Versatile Disc)

Capacity: 4.7 GiB (single layer) to 17 GiB (dual layer, double sided)
Uses: Movies, software distribution, backups
DVD-R, DVD+R, DVD-RW, DVD+RW

Blu-ray

Capacity: 25 GiB (single layer) to 128 GiB (quad layer)
Uses: HD/4K movies, games, large archives
BD-R (write once), BD-RE (rewritable)

How optical storage works:

- A laser reads/writes data by detecting pits and lands on the disc surface
- Pits are small indentations that scatter light; lands reflect light
- The pattern of reflections is interpreted as binary data
- Blu-ray uses a shorter wavelength (blue) laser, allowing more data to be stored

Characteristics:

- **Portable:** Thin, lightweight discs
- **Cheap to produce:** Low cost for mass distribution
- **Slow access:** Spinning disc and laser positioning takes time
- **Susceptible to scratches:** Physical damage can make data unreadable
- **Limited capacity:** Much less than HDDs or SSDs

> **Note:** Optical storage is less common today due to streaming services and cloud storage, but is still used for software distribution, archiving, and situations where physical media is required.

Worked Example 5 — Converting storage units

Question: A file is 5,242,880 bytes in size. Express this file size in MiB. Show your working. [2 marks]

Step 1: Convert bytes to KiB: 5,242,880 ÷ 1024 = 5,120 KiB **(1)**
Step 2: Convert KiB to MiB: 5,120 ÷ 1024 = 5 MiB **(1)**
Alternative method: 5,242,880 ÷ 1024^2 = 5,242,880 ÷ 1,048,576 = 5 MiB

Exam tip: To convert bytes → KiB → MiB → GiB, divide by 1024 at each step. Recognise that 1024^2 = 1,048,576 and 1024^3 = 1,073,741,824.

Worked Example 6 — Comparing storage capacities

Question: An HDD has a capacity of 2 TiB. A Blu-ray disc can store 25 GiB. Calculate how many Blu-ray discs would be needed to back up the entire HDD. [2 marks]

Step 1: Convert HDD to GiB: 2 × 1024 = 2048 GiB **(1)**
Step 2: Divide by Blu-ray capacity: 2048 ÷ 25 = 81.92, round up = 82 discs **(1)**
Must round up because 81 discs wouldn't be enough

Exam tip: When calculating "how many needed to fit", always round UP. You can't use a fraction of a disc, and rounding down would leave data without backup.

Worked Example 7 — SSD vs HDD comparison

Question: A company is upgrading their laptops. Give two reasons why they might choose SSDs instead of HDDs. [2 marks]

Model answer:
SSDs are faster than HDDs because they have no moving parts, so programs and files load more quickly **(1)**.
SSDs are more durable than HDDs because they have no moving parts, making them more resistant to damage from being moved around **(1)**.
Also acceptable: SSDs are lighter/quieter/use less power

Exam tip: Always link the advantage to the context. Laptops are **portable**, so durability and weight matter more than for a desktop.

Worked Example 8 — Optical storage capacity calculation

Question: A software company wants to distribute a program that is 3.2 GiB in size. State whether a single DVD (4.7 GiB capacity) or multiple CDs (700 MiB each) would be more practical, and calculate how many CDs would be needed. [3 marks]

Step 1: Compare to DVD: 3.2 GiB < 4.7 GiB, so one DVD would fit **(1)**
Step 2: Convert to MiB for CD calculation: 3.2 × 1024 = 3,276.8 MiB **(1)**

Step 3: Calculate CDs needed: 3,276.8 ÷ 700 = 4.68, round up = 5 CDs **(1)**
Conclusion: One DVD is more practical than 5 CDs.

Exam tip: Exam questions often ask you to compare options. Show the calculation for both, then state which is more practical and why.

3. Choosing appropriate storage

Exam questions often ask you to recommend or justify a storage choice for a specific scenario. To answer these well, you need to understand the six key characteristics and how they apply to different situations.

3.1 The six storage characteristics

When comparing storage devices, consider these six characteristics:

1. Capacity

How much data can be stored.
Highest: HDD (up to 20+ TiB)
Lowest: Optical (700 MiB — 128 GiB)

2. Speed

How quickly data can be read/written.
Fastest: SSD
Slowest: Optical

3. Portability

How easily the device can be moved.
Most portable: USB flash drive, memory card
Least portable: Internal HDD

4. Durability

How well the device resists physical damage.
Most durable: SSD (no moving parts)
Least durable: HDD (fragile mechanical parts)

5. Reliability

How consistently the device works over time.
Most reliable: SSD (for normal use)
Concerns: Optical (scratches), HDD (mechanical failure)

6. Cost

Price per unit of storage.
Cheapest per GiB: HDD
Most expensive per GiB: SSD

3.2 Matching storage to scenarios

Scenario	Recommended storage	Key reasons
Gaming laptop boot drive	SSD	Fast loading times, durable for portable use

Scenario	Recommended storage	Key reasons
Home backup of photos and videos	External HDD	High capacity, low cost for large media files
Distributing software to customers	Optical (DVD/Blu-ray)	Cheap to mass produce, portable, no internet needed
Transferring files between school computers	USB flash drive	Highly portable, plug-and-play, rewritable
Data centre storing customer records	HDD arrays (or SSD for high-access data)	Massive capacity needed, cost-effective for bulk storage
Smartphone internal storage	Solid-state (embedded flash)	Small size, low power, durable, no moving parts

3.3 Storage calculations

Exam questions may ask you to calculate how much storage is needed or how many files can fit on a device. Remember to use powers of 1024 for conversions.

Common calculation patterns:
Total storage needed: Number of files × size of each file
Number of files that fit: Total capacity ÷ size of each file
Converting bytes to larger units: Divide by 1024 repeatedly
Converting larger units to bytes: Multiply by 1024 repeatedly

Worked Example 9 — Justifying storage choice

Question: A school is replacing the storage in its computer suite. The computers are used for programming, web browsing, and word processing. Explain why SSDs would be more suitable than HDDs for this scenario. [4 marks]

Model answer:
Speed: SSDs are faster than HDDs, so programs will load more quickly, improving productivity **(1)**.
Durability: SSDs have no moving parts, so they are less likely to be damaged by students accidentally bumping the desks **(1)**.
Reliability: SSDs are more reliable for frequent read/write operations like saving work, reducing the risk of data loss **(1)**.
Noise: SSDs are silent, creating a quieter working environment for the classroom **(1)**.

Exam tip: For 4-mark justify questions, give 4 distinct points. Each point should name a characteristic (speed, durability, etc.) and link it to the scenario

(school, students, programming, etc.).

Worked Example 10 — Calculating storage requirements

Question: A photographer takes 500 photos. Each photo is 6 MiB in size. Calculate the total storage needed in GiB. [2 marks]

Step 1: Calculate total in MiB: 500 × 6 = 3000 MiB **(1)**
Step 2: Convert to GiB: 3000 ÷ 1024 = 2.93 GiB (or 3000/1024 GiB) **(1)**
Accept 2.93 GiB, approximately 3 GiB, or the expression 3000/1024

Exam tip: Show your working! Even if your final answer is slightly off, you can get marks for the correct method. Always divide by 1024 (not 1000) for binary units.

Worked Example 11 — How many files fit?

Question: A USB flash drive has a capacity of 16 GiB. A student wants to store video files that are each 512 MiB. Calculate how many complete video files can be stored. [2 marks]

Step 1: Convert 16 GiB to MiB: 16 × 1024 = 16,384 MiB **(1)**
Step 2: Divide by file size: 16,384 ÷ 512 = 32 files **(1)**

Exam tip: "Complete files" means you must round down if you get a decimal. You can't store part of a file, so 32.5 would become 32.

Worked Example 12 — Extended comparison question

Question: A video production company needs to choose between an HDD and an SSD for editing 4K video footage. The footage files are very large. Discuss the advantages and disadvantages of each option. [6 marks]

Model answer:
HDD advantages:
HDDs have higher maximum capacities (up to 20+ TiB), which is beneficial for storing very large 4K video files **(1)**.
HDDs are cheaper per GiB, making them more cost-effective for storing large amounts of footage **(1)**.
HDD disadvantages:

HDDs are slower due to mechanical read/write heads, which may cause delays when editing large video files **(1)**.
SSD advantages:
SSDs are much faster, which speeds up loading, scrubbing through footage, and rendering **(1)**.
SSDs are more reliable for constant read/write operations during editing **(1)**.
SSD disadvantages:
SSDs have a higher cost per GiB, which increases the budget for large storage requirements **(1)**.
Conclusion: A hybrid approach (SSD for active editing, HDD for archive storage) may offer the best balance.

Exam tip: For "discuss" questions, present both sides. Structure your answer clearly with advantages and disadvantages for each option. A brief conclusion shows balanced judgement.

Worked examples summary

The table below summarises the 12 worked examples covered in this topic, organised by section and difficulty level.

#	Topic	Question type	Marks	Key skill
1	RAM vs ROM	State difference	2	Paired comparison (volatile/non-volatile)
2	RAM requirements	Calculate total RAM	2	Add requirements, convert units
3	Volatility	Explain scenario	2	Link concept to real situation
4	ROM boot process	Explain role	2	Describe boot sequence
5	Storage units	Convert bytes to MiB	2	Divide by 1024 chain
6	Storage comparison	Calculate discs needed	2	Convert units, divide, round up
7	SSD vs HDD	Give reasons	2	Justify choice with context
8	Optical capacity	Calculate and compare	3	Compare DVD vs multiple CDs
9	Storage choice	Explain suitability (extended)	4	Multiple characteristics + context

#	Topic	Question type	Marks	Key skill
10	Storage calculation	Calculate total storage	2	Multiply then convert units
11	Storage calculation	Calculate number of files	2	Convert then divide
12	HDD vs SSD	Discuss advantages/disadvantages	6	Balanced extended response

Quick reference

Key terms and definitions

Term	Definition (mark-scheme style)
Primary memory	Memory that the CPU can access directly; includes RAM and ROM
Secondary storage	Non-volatile storage used for long-term data storage; includes HDDs, SSDs, optical discs
RAM	Random Access Memory; volatile memory that stores data and programs currently in use
ROM	Read-Only Memory; non-volatile memory that stores permanent boot instructions (BIOS/UEFI)
Volatile	Memory that loses its contents when the power is turned off
Non-volatile	Memory/storage that retains its contents when the power is turned off
HDD	Hard Disk Drive; magnetic storage using spinning platters and a read/write head
SSD	Solid-State Drive; storage using flash memory with no moving parts
Optical storage	Storage that uses a laser to read/write data on a disc (CD, DVD, Blu-ray)
Flash memory	Non-volatile storage that uses electrical charges in memory cells; found in SSDs and USB drives
Capacity	The amount of data a storage device can hold, measured in bytes (KiB, MiB, GiB, TiB)
BIOS/UEFI	Firmware stored in ROM that initialises hardware and loads the operating system on startup

Storage units reference

Unit	Bytes	Power of 2	Practical reference
1 KiB	1,024	2^{10}	A short text document
1 MiB	1,048,576	2^{20}	A high-resolution photo
1 GiB	1,073,741,824	2^{30}	A short HD video or a few hundred songs
1 TiB	1,099,511,627,776	2^{40}	Hundreds of HD movies or a large game library

Storage comparison at a glance

Type	Technology	Speed	Capacity	Cost/GiB	Durability	Best for
RAM	Volatile chips	Fastest	4—64 GiB	Highest	N/A	Active processing
SSD	Flash memory	Very fast	Up to ~8 TiB	High	High	Boot drive laptops
HDD	Magnetic platters	Moderate	Up to 20+ TiB	Low	Low	Bulk storage, backups
USB flash	Flash memory	Fast	Up to ~2 TiB	Medium	High	File transfer, portable
Optical	Laser on disc	Slow	700 MiB — 128 GiB	Very low	Medium	Distributio archives

Typical pitfalls

Pitfall 1 — Confusing RAM and ROM.
Wrong: "ROM stores the operating system while it runs."
Right: RAM stores the operating system while it runs. **ROM** stores the boot instructions (BIOS) that load the OS at startup.

Pitfall 2 — Saying RAM is non-volatile.
Wrong: "RAM keeps data when the power is off."
Right: RAM is **volatile** — it loses all data when the power is turned off. Only ROM and secondary storage are non-volatile.

Pitfall 3 — Confusing primary and secondary.
Wrong: "An SSD is a type of primary memory."
Right: SSDs are **secondary storage**. Primary memory is RAM and ROM (directly accessible by CPU). Secondary storage requires loading into RAM first.

Pitfall 4 — Using wrong conversion factor.
Wrong: "1 GiB = 1000 MiB" or dividing by 1000.
Right: Always use **1024** for binary units: 1 GiB = 1024 MiB, 1 MiB = 1024 KiB.

Pitfall 5 — Saying HDDs are faster than SSDs.
Wrong: "HDDs are faster because they have been around longer."
Right: SSDs are **much faster** than HDDs because they have no moving parts. HDDs are limited by mechanical seek time.

Pitfall 6 — Forgetting why SSDs are more durable.
Wrong: "SSDs are more durable because they are newer technology."
Right: SSDs are more durable because they have **no moving parts**, so they are not damaged by shocks or vibration. HDDs have fragile spinning platters and read/write heads.

Pitfall 7 — Mixing up capacity rankings.
Wrong: "CDs can store more than HDDs."
Right: HDDs have the **highest capacity** (up to 20+ TiB). SSDs are next (up to ~8 TiB). Optical storage has the **lowest capacity** (CDs: ~700 MiB).

Pitfall 8 — Ignoring context in justify questions.
Wrong: "SSDs are better because they are faster." (No link to scenario)
Right: Link to the scenario: "SSDs are better **for the laptop** because they are faster, so **the user** can start working more quickly."

Pitfall 9 — Forgetting ROM's role in boot process.
Wrong: "The operating system is stored in ROM."
Right: ROM stores the **BIOS/boot instructions**, which then load the operating system from secondary storage into RAM.

Pitfall 10 — Saying optical storage is obsolete.
Wrong: "Optical storage is never used anymore."
Right: Optical storage is still used for **software distribution, archiving, and physical media sales** (games, movies). It's less common but not obsolete.

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